

WASP-104b

R-2100

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_250513 · created 2026-08-10T23:18:29+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460808.6602 +/- 0.0063 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.152 +/- 0.045
Transit depth (Rp/Rs)^2	2.3 +/- 1.36 [%]
Orbital Inclination (inc)	83.33 +/- 2.8
Airmass coefficient 1 (a1)	0.51 +/- 0.14
Airmass coefficient 2 (a2)	0.5 +/- 0.15
Scatter in the residuals of the lightcurve fit is	23.73 %
Best Comparison Star	None
Optimal Aperture	2.76
Optimal Annulus	6.75
Transit Duration (day)	0.037 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.152 $\pm$ 0.045 vs 0.1214 (deviation 0.41 σ)
Duration fit vs published	0.037 d vs 0.0742 d (deviation 1.32 σ)
Mid-time O-C	-13.4 min
Depth SNR	2.0
Residual scatter	23.73 %

Light curve

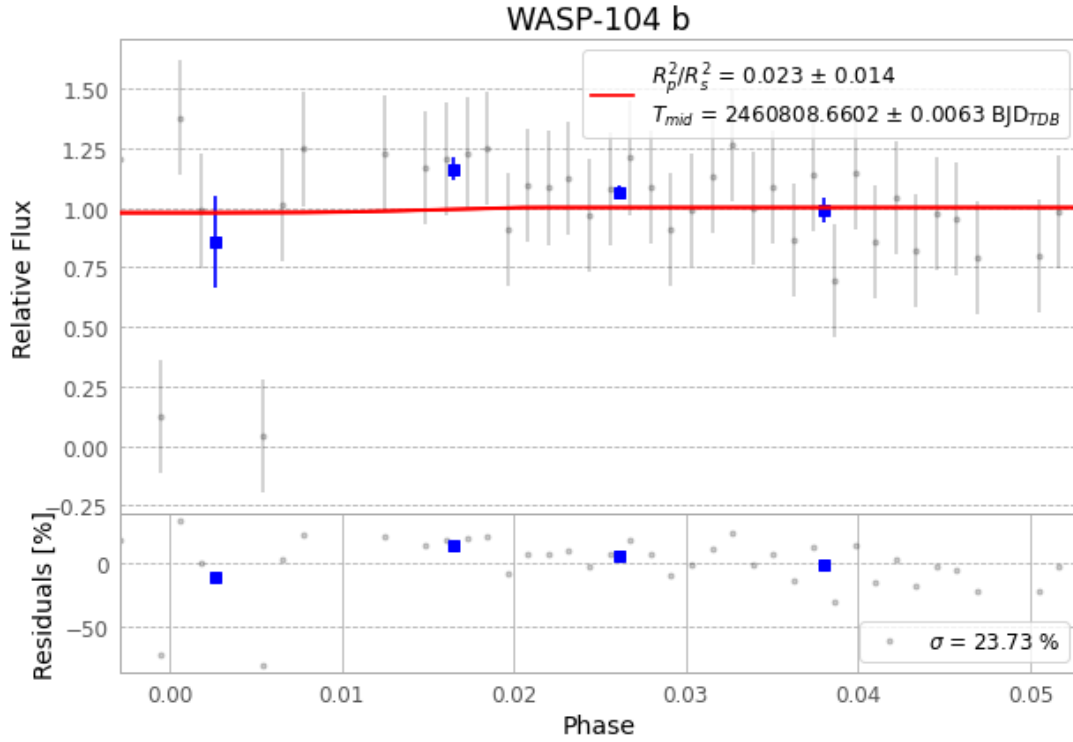


Figure 1 — FinalLightCurve_WASP-104 b_2025-05-12.png (SHA-256 99a06a9674798f54...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [259, 123], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d5d07698fb0c6d0c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

47 FITS frames (31 MB), WASP-104250513033913.FITS → WASP-104250513060015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-250513.log (9032709a496b...), FinalLightCurve_WASP-104 b_2025-05-12.png (99a06a967479...), FinalLightCurve_WASP-104 b_2025-05-12.csv (d7de7670a32a...)

Compute resources

Wall time 212.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-10 23:18Z.